magneto

_travelers guide
_for mastering magnetism

4	3	2
playback heads	modes	min of tape

[tap] press **[tap]** hold › then noon = 12 o'clock cw / ccw = clockwise / counter

four playback heads, one record head · echo, looper, phrase sampler, spring reverb, oscillator

magneto · 28 hp · manual rev b

cyan = hold

01 buttons · LEDs

mg_01

[modes]	echo ● › loop ● › sample ● › echo …
[tap] ×2	delay time at head 4, 50 ms–15 s · loop / sample: splice in › out › clear (04)
[1]–**[4]**	● green = that head feeds back into the record head, even with its level at zero
[transport]	the four buttons turn ● red and become ∞ ◂▸ ▸‖ ‖ (06)

02 knobs

mg_02

[speed / pitch]	tape speed, 8:1 · min ¼ · about 9 o'clock ½ · noon normal · max 2× (fig.04)
[dry]	your input to the outputs
[wet]	all four heads to the outputs, after their levels
[rec lvl]	into the record head · clean to fat and saturated
[repeats]	feedback for every head whose button is green
[level 1–4]	each head's volume · **[** ● **n]** + turn its knob = pan it (pan on center)

03 tape mechanics

mg_03

[low cut]	min = full lows · max = thin, peaky "magnetic drum" repeats
[tape age]	min = fresh tape · cw = warmer · max = a peak that blooms at high repeats
[crinkle]	friction, creases, splices, dirt · min clean · max chewed
[wow & flutter]	ccw = a tuned machine · cw = one that needs service
[spring]	spring reverb level · dry and wet both feed it

crinkle and wow & flutter bite much harder at slow speeds

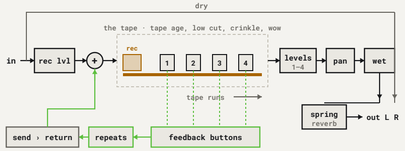
05 switches

mg_05

[heads]	**even** 1/16 · 1/8 · dotted 1/8 · 1/4 · **triplet** 1/16t · 1/8t · 1/4t · 1/4 · **shift** pitched heads (fig.02)
[pan]	**LRLR** 1 3 left · 2 4 right · **center** all middle, custom pans · **LRRL** 1 4 left
one output	everything sums to mono at that jack

__fig.01 : signal path

mono in, stereo out



only heads with a green button feed back, even with their level at zero
loop / sample: head 4 sits before the record head and plays the loop

04 modes

mg_04 · **[modes]**

● echo	**four-head tape delay** · **[tap]** sets the head spacing, **[speed]** the tape speed
● loop	**sound-on-sound** · head 4 plays the loop, heads 1–3 keep echoing · needs **[4]** green and level 4 up
loop **[tap]**	splice in › splice out (plays) › clear · **[repeats]** lower = the loop fades as you layer
● sample	**phrase sampler** · **[tap]** record › stop › clear · **[▸|]** plays it once
length	longest loop or sample: 15 s at max speed · 30 s noon · 1 min half · 2 min min
heads 1–3	keep the delay time you had when you entered loop / sample, or follow **[clk in]**

__fig.04 : the speed dial

8:1



slower
longer, lower, darker · wow and crinkle grow

faster
shorter, higher, cleaner

tap instead
moves the heads: new delay, same pitch

turning it
repitches what's on the tape

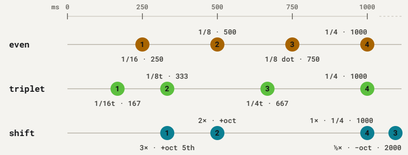
09 audio

mg_09 · 20 vpp

[left / right in]	left alone = mono, dry still reaches both outs
[left / right out]	one cable = everything summed to mono
[send] **[return]**	an insert before the record head · both patched = repeats mono, dry stays stereo
the module	28 hp · 41 mm deep · 210 mA on each rail · 96 khz · delays from 200 µs to 2 min

__fig.02 : head spacing

and the clk outs



tapped or clocked at 1000 ms · clk 1–4 out tick at these times, 50% duty, phase-aligned
shift = each head plays at its own speed · loop / sample: clk 4 out = the loop length

07 pitch quantize

mg_07 · **[tap]**

[tap]	**[speed / pitch]** snaps to the notes of a scale · tap LED ●
[transport] + **[speed]**	turn to pick one of 15 scales · buttons 1–4 show which (fig.05)
range	two octaves (4:1) · chromatic 1.2 octaves · speed cv snaps to half steps

__fig.05 : the 15 scales

[transport] + **[speed]**

	1	2	3	4
pentatonic	○	○	○	●
minor blues	○	○	●	○
major	○	○	●	●
dorian	○	●	○	○
phrygian	○	●	○	●
lydian	○	●	●	○
mixolydian	○	●	●	●
aeolian	●	○	○	○
locrian	●	○	○	●
harmonic minor 5th	●	○	●	○
harmonic minor	●	○	●	●
whole tone	●	●	○	○
major in octave jumps	●	●	○	●
harmonic series	●	●	●	○
chromatic (1.2 oct)	●	●	●	●

buttons 1–4
count in binary:
button 4 = 1,
3 = 2, 2 = 4, 1 = 8

hold transport
+ turn speed

08 self-oscillation

mg_08 · no input needed

1. **[speed / pitch]** fully ccw · **[repeats]** at 100%
2. **[tap]** ×2, as fast as you can · it starts to sing
3. play **[speed]** and the four tape knobs · it wanders through pitches
4. **[transport]** › **[∞]** locks the pitch · speed and speed cv still play it

speed cv is 1 v/oct, −3 to +3 v: six octaves, three below the knob alone · LRLR / LRRL widen the tone

12 dip switches

mg_12 · on the back, off by default

s1 **[feedback cv]**	on: with the transport LED off, the transport jacks toggle feedback buttons 1–4 instead
s2 **[dual split]**	on: left in / out = mono four-head delay · right in / out = mono spring reverb

06 transport

mg_06 · **[transport]** first

[∞] infinite	record head off · the last delay cycle or loop plays forever
[◂▸] reverse	from that moment · echo: the whole tape backwards · loop / sample: the loop backwards
[▸|] restart	loop / sample from the top · realigns shift heads · leaves pause instantly
[‖] pause	the tape slows to a stop, or spins back up · while paused, level 4 = how fast
scrub	while paused, **[speed / pitch]** scrubs about a foot of tape · slower speed = more audio
defaults	all latch except ▸|, always a trigger · change one at power on (mg_15)

10 cv inputs

mg_10

continuous, −5 to +5 v — the knob is the offset

[speed]	−3 to +3 v, 1 v/oct
[wet] **[repeats]** **[spring]**	add to their knobs

triggers, 0–5 v rising edge

[clk in]	delay time as a quarter note, 50 ms–15 s
[rec gate]	record head on / off · rec lvl LED ● when muted
[shift]	shift head mode on / off
[∞] **[◂▸]** **[▸|]** **[‖]**	the transport buttons
[tap]	the tap button: delay, splice or sample · 10 ms–15 s

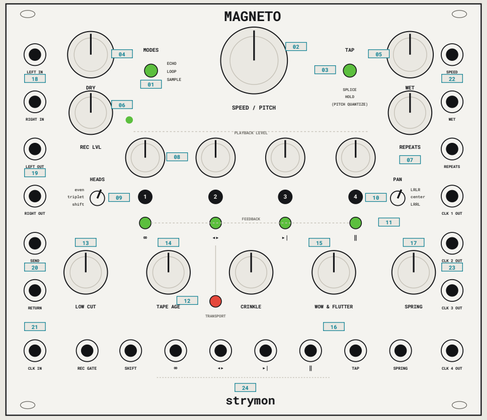
11 tricks

mg_11 · from the manual

sub-octave	shift heads · a gate into **[▸|]** cv = a zero-latency sub-octave from head 3
endless	sample mode · **[clk 4 out]** › **[▸|]** cv retriggers the phrase
extra tone	**[∞]** on a very short delay adds a harmonic at the seam

__fig.03 : front panel

01–17 = controls · 18–24 = jacks



[01]	**__modes :** echo · loop · sample	
[02]	**__speed / pitch :** tape speed 8:1	
[03]	**__tap :** delay · splice · hold = quantize	
[04]	**__dry :** input level	
[05]	**__wet :** all heads	
[06]	**__rec lvl :** into the tape · LED	
[07]	**__repeats :** feedback	
[08]	**__level 1–4 :** each head	
[09]	**__heads :** even · triplet · shift	
[10]	**__pan :** LRLR · center · LRRL	
[11]	**__1–4 :** feedback · transport	
[12]	**__transport :** flips 11 to red	
[13]	**__low cut :** thins the repeats	
[14]	**__tape age :** darker, peakier	
[15]	**__crinkle :** damaged tape	
[16]	**__wow & flutter :** speed wobble	
[17]	**__spring :** reverb level	
[18]	**__ins :** left = mono	
[19]	**__outs :** one = mono sum	
[20]	**__send** · **return :** before record	
[21]	**__clk in :** quarter note	
[22]	**__cv :** speed · wet · repeats	
[23]	**__clk 1–4 out :** the heads	
[24]	**__gates :** rec · shift · transport · tap	

_where things hide

no menus · mg_15

held buttons

- **tap** pitch quantize
- **transport + speed** pick a scale
- **green n + level n** pan head n (pan center)
- **paused: level 4** slowdown speed

at power on

- **hold ∞ ◂▸ ▸\| or ‖** flips it: momentary ● / latch ●
- **dip switches** s1 feedback cv · s2 dual split (12)

LED codes

modes · tap	● echo ● loop ● sample ● tap = quantize
buttons 1–4	● feedback on ● transport layer / scale code
rec lvl	● none ● clean ● saturating ● heavy